

1. Direct proof

Prove that the sum of two even integers is even.

Let a and b be even integers.

Then $a = 2m$ and $b = 2n$ for some integers m and n .

$$\begin{aligned} \text{So } a + b &= 2m + 2n \\ &= 2(m + n). \end{aligned}$$

Because $m + n$ is an integer, $a + b$ is even.

This completes the proof.

2. Mathematical induction

Prove that $1 + 2 + \dots + n = n(n + 1) / 2$
for every positive integer n .

Base case: $n = 1$.

$$1 = 1(1 + 1) / 2 = 1.$$

Suppose the formula holds for k .

$$1 + 2 + \dots + k = k(k + 1) / 2.$$

For $k + 1$, assume that

$$1 + 2 + \dots + k + (k + 1) = (k + 1)(k + 2) / 2.$$

Therefore the formula holds for every positive integer n .

3. Sets & notation

Express the positive even integers in set-builder notation.
State the domain explicitly.

The positive even integers are 2, 4, 6, 8, ...

My set-builder notation:

$$E = \{ 2k \mid k > 0 \}$$

This generates the even numbers greater than zero.